

Self-FEP Memory: A Precision-Weighted Generative Self Drives Memory Encoding, Forgetting, and Retrieval

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Abstract

Forty years of work on the Self-Reference Effect (SRE) has established a robust empirical pattern — self-referenced encoding produces better recall than semantic, phonemic, or structural encoding — without a mechanistic computational account that connects the effect to the Free Energy Principle (FEP). We introduce **Self-FEP Memory**, a mechanism in which the agent’s self is a generative model $M_{\text{self}} = (\mu_{\text{self}}, \pi_{\text{self}})$ with running embedding centroid μ_{self} and a precision-inspired gating term π_{self} . Self-precision multiplicatively gates encoding depth, forget score, and retrieval ranking through a single self-relevance signal. We instantiate the mechanism as a ~ 200 -line extension to an existing cognitive-layer substrate and run three controlled CPU experiments: (E1) the forget-score formula has the qualitative response to controlled self-relevance inputs predicted by depth-of-processing theory, in 10/10 seeds; (E2) a 2×2 Mood $\times$ Self factorial recovers the multiplicative cell-mean structure analytically expected from a product-of-resistances formula and rejected by an additive linear model; (E3) a true π_{self} ablation monotonically attenuates the SRE size from 0.085 at $\pi=1$ to 0.000 at $\pi=0$, consistent with empirical reports of SRE attenuation in dissociation and severe depression. We name five scope boundaries (section 6) and a corresponding validation roadmap; the present work establishes the mechanism, not the quantitative human-data fit. The substrate is open-source and the experiments complete in seconds on a single CPU.

1 Introduction

Information encoded with reference to the self is remembered better than information processed for meaning, sound, or surface form. This pattern, the *Self-Reference Effect* (SRE), has been documented since Rogers, Kuiper, and Kirker [Rogers et al.(1977)Rogers, Kuiper, and Kirker] and confirmed in the meta-analysis of Symons and Johnson [Symons and Johnson(1997)] with a robust Cohen’s $d \approx 0.50$ separation between self-referenced and semantically encoded items. The effect survives across age groups, languages, source-memory paradigms, and clinical attenuations [Conway(2005)].

What computational mechanism produces it? Existing models stop short. Retrieved-context theories of episodic memory — TCM [Howard and Kahana(2002)], CMR [Polyn et al.(2009)Polyn, Norman, and Kahana], and the emotional extension CMR3 [Cohen and Kahana(2022)] — model emotion-modulated recall but lack a representation of self. Activation-based theories like ACT-R [Anderson(2007)] treat all chunks symmetrically. Retrieval-augmented language-model agents [Sumers et al.(2024)Sumers, Yao, Narasimhan, and Griffiths, Memtensor Lab(2025)] use embedding similarity plus heuristic decay but do not distinguish self-relevance from topical relevance. Conversely, the Free Energy Principle (FEP) literature on self [Friston(2010), Apps and Tsakiris(2014), Limanowski and Blankenburg(2013)] articulates a generative model of selfhood but does not connect it to memory dynamics: precision-weighted self priors have been linked to body recognition and minimal phenomenal selfhood, not to encoding depth or forgetting rate.

The white space is the intersection. We propose to model the self as a generative model $\mathcal{M}_{\text{self}} = (\mu_{\text{self}}, \pi_{\text{self}})$ under the FEP and to let its precision multiplicatively gate memory:

$$\text{forget_score} = \underbrace{\text{recency} \cdot \text{usage}}_{\text{Ebbinghaus}} \cdot \frac{1}{1 + \text{importance}} \cdot \frac{1}{1 + \alpha|V|A} \cdot \frac{1}{1 + \lambda \text{sr}} \quad (1)$$

where $\text{sr} = \pi_{\text{self}} \cdot \frac{1 + \cos(e_i, \mu_{\text{self}})}{2} + (1 - \pi_{\text{self}}) \cdot 0.5$ is the self-relevance of memory i given its embedding e_i . The self prior μ_{self} is updated by exponential moving average toward *identity-constitutive* utterances — user content containing first-person tokens — gated by emotional significance. The precision π_{self} follows an FEP-inspired rule $\pi_{\text{self}} \propto 1/(1 + \gamma \text{Var}[\text{PE}])$ over a rolling prediction-error window.

The contributions of this paper are:

1. **A precision-weighted generative self that drives memory dynamics (section 3).** The self enters the existing `forget_score` formula as a multiplicative factor, dual to the emotional intensity factor; entering it into retrieval ranking adds a single weighted similarity term.
2. **Controlled self-relevance response (E1, section 4).** On a 4-condition encoding paradigm with condition-typed self-relevance values drawn from depth-of-processing theory, the forget-score ordering follows the human SRE direction in 10/10 seeds. We are explicit that sr is manually assigned, so this tests the gating formula’s qualitative response — not the stimulus-to-memory forward pass.
3. **Non-parallel raw-scale cell means (E2, section 4).** A 2×2 factorial recovers the multiplicative structure analytically expected from eq. (5): cell-mean differentials are not constant (diagnostic magnitude 0.71), so the raw-scale linear-additive model is rejected against the model containing the interaction term. We present the analytic interpretation alongside the AIC value, because the AIC magnitude itself is fragile under deterministic data.
4. **Monotone π_{self} attenuation (E3, section 4).** A true precision ablation through the `SelfModel` forward pass yields SRE size 0.000 at $\pi_{\text{self}}=0$, growing monotonically to 0.085 at $\pi_{\text{self}}=1$. This is consistent with empirical SRE attenuation reported in dissociation and severe depression.
5. **Open-source substrate.** The complete implementation is a ~ 200 -line extension to `BorgeAgent` [Labs(2026)] and ships as `v0.2-cleaner`; all experiments are CPU-only and run in seconds.

Section 6 states five scope boundaries together with the validation roadmap they imply: a bag-of-tokens embedding that defers a true raw-stimulus SRE forward pass (L1) to follow-up with a sentence-transformer encoder; a precision-inspired gating rule that is not yet derived from a variational objective (L2); the absence of a quantitative fit to human meta-analytic effect sizes (L3). The present work establishes the mechanism and its falsifiable predictions; the empirical quantitative fit is the explicit next step.

2 Related Work

Retrieved-context memory models. The Temporal Context Model [Howard and Kahana(2002)] and its descendants [Polyn et al.(2009)Polyn, Norman, and Kahana, Cohen and Kahana(2022)] represent each studied item as a feature vector bound to a drifting context vector; retrieval reinstates context, which probabilistically reactivates associated items. CMR3 [Cohen and Kahana(2022)] extends CMR with a bi-cell representation of emotional valence and predicts mood-congruent recall, intrusive memories, and the effects of emotion dysregulation. These models do not include a self representation: the context vector drifts with topic, not identity, and emotional features are content-attached, not agent-attached.

Activation-based memory. ACT-R [Anderson(2007)] stores declarative chunks with base-level activation $B_i = \ln \sum_j t_j^{-d}$ summing over past retrievals. Recent LLM-agent memory architectures port ACT-R-style activation to dialog memory [Memtensor Lab(2025)], combined with surprise or emotional arousal for retention; a recent wave of 2025 systems carry this style further with cognitive-science-inspired organisation. None of these treat the self as a distinct generative entity that gates encoding.

FEP and self. Friston’s free-energy principle [Friston(2010), Friston et al.(2017)] formalises perception and action as minimising variational free energy under a generative model. *Self-evidencing* interpretations [Hohwy(2016), Seth and Tsakiris(2018)] cast the agent as actively sampling sensations that confirm its model of itself. Apps and Tsakiris [Apps and Tsakiris(2014)] apply this to bodily self-recognition; Limanowski and Blankenburg [Limanowski and Blankenburg(2013)] map minimal phenomenal selfhood to a hierarchical generative model. The thread we pick up is the implicit one: if the self is a generative model with precision π_{self} , that precision should modulate memory the same way it modulates other generative-model-driven processes.

Self-Reference Effect. Rogers, Kuiper, and Kirker [Rogers et al.(1977)] showed that self-referenced encoding produces a robust advantage over semantic, phonemic, and structural encoding. Symons and Johnson [Symons and Johnson(1997)] synthesised the meta-analysis. Recent work confirms the SRE in older adults, source-memory paradigms, and cross-cultural samples [Conway(2005), Klein(2012)]. The empirical pattern is exceptionally well-documented; what is missing is a computational mechanism that connects it to a broader theory of mind.

Encoding and forgetting. Levels of processing [Craik and Lockhart(1972)] and encoding specificity [Tulving and Thomson(1973)] predict that depth of processing and context match drive recall. Ebbinghaus [Ebbinghaus(1885)] quantified forgetting as a near-power-law of time. Bower [Bower(1981)] established the associative-network account of mood-congruent memory. Nader and Hardt [Nader and Hardt(2009)] reviewed reconsolidation, the phenomenon that retrieved memories become labile and can be re-encoded. Our model builds on these foundations but unifies them with the FEP self, which none of the prior work explicitly formalises in a memory context.

Position of this work. We are not claiming the first computational model of the SRE, nor the first FEP account of self, nor the first emotion-modulated memory model. We are claiming the first model that wires *all three* together into a single mechanism whose multiplicative interaction structure generates falsifiable predictions, and that can be tested as a thin extension to an existing agent substrate. The closest comparison target, CMR3, is emotion-only and lacks a self dimension; we leave a quantitative side-by-side comparison to future work (section 6).

3 Method

Scope: precision-inspired, not variational. Before describing the mechanism, we make the FEP scope precise. We use the FEP as a *design discipline*: the self is treated as a generative model with a precision-weighted prior, and self-precision gates downstream processing the way precision does throughout the FEP literature. We do *not* derive the precision update rule from a variational free-energy objective; eq. (3) is engineered to have the qualitative shape FEP precision should have (low PE variance $\Rightarrow$ high precision) and is a first commitment, not a theorem. A variational derivation under a specified likelihood-prior pair is named explicitly as a follow-up step (section 6, L2). Where the paper says “FEP-inspired” or “FEP-grounded mechanism,” it means: faithful to the precision-gating intuition; not yet faithful to variational free-energy minimisation as an objective.

3.1 The Self as a Generative Model

We treat the agent’s self as a one-component generative model $\mathcal{M}_{\text{self}} = (\mu_{\text{self}}, \pi_{\text{self}})$ over an embedding space $\mathbb{R}^d$. The mean $\mu_{\text{self}} \in \mathbb{R}^d$ is the running centroid of identity-constitutive content; the scalar precision $\pi_{\text{self}} \in [0, 1]$ encodes the agent’s confidence in μ_{self} . We use $d = 64$ throughout.

Embedding. For an utterance u tokenised as $\{t_1, \dots, t_n\}$, we use a deterministic SHA1-bag embedding $e(u) = \text{normalise}(\sum_{i=1}^n \text{hash}_d(t_i))$, where each token hashes to a unit-vector contribution per dimension and the sum is L^2 -normalised. This embedding has zero external dependencies and is fully deterministic, but cannot separate framing words from content words — a limitation we discuss in section 6 (L1).

Self-token gating. μ_{self} updates only on user utterances that contain first-person tokens (English: *i, me, my, myself, ...*; Chinese first-person markers *wo, ziji, ...*). This is the cognitive-science commitment that the self is identity-constitutive rather than topic-tracking: utterances about arbitrary content do not reshape the self prior.

Update rule. Given a self-referencing user utterance u with emotional significance $w_u = |V_u| \cdot A_u$, the update is exponential-moving-average:

$$\mu_{\text{self}} \leftarrow (1 - \alpha w_u) \mu_{\text{self}} + \alpha w_u e(u), \quad \alpha = 0.1, \quad (2)$$

with prediction error $\text{PE} = \|\mu_{\text{self}}^{\text{new}} - \mu_{\text{self}}^{\text{old}}\|$ appended to a rolling window of size $W=50$.

Precision rule. Precision is FEP-inspired:

$$\pi_{\text{self}} \propto \frac{1}{1 + \gamma \text{Var}[\text{PE}_t]_{t \in W}}, \quad \gamma = 10. \quad (3)$$

Low PE-variance (the self prior reliably predicts incoming self-relevant content) yields high precision. High PE-variance (unstable self-evidence) yields low precision. We are explicit that this rule is engineered rather than derived from a full variational objective (section 6, L2); the qualitative behaviour — stability $\Rightarrow$ high precision — matches the FEP literature.

3.2 Self-Relevance

Given memory i with embedding e_i , its self-relevance is the precision-weighted cosine similarity to μ_{self} :

$$\text{sr}_i = \pi_{\text{self}} \cdot \frac{1 + \cos(e_i, \mu_{\text{self}})}{2} + (1 - \pi_{\text{self}}) \cdot 0.5. \quad (4)$$

At $\pi_{\text{self}} = 1$ the score is the rescaled cosine. At $\pi_{\text{self}} = 0$ it collapses to 0.5 (neutral, no self prior) — which is the formal mechanism by which low precision attenuates the SRE (E3a).

3.3 Memory Dynamics

We extend an existing `forget_score` formula (recency / usage / importance / emotion-resistance) with a self-resistance term:

$$\text{forget_score}_i = \underbrace{d_i^{0.7}}_{\text{Ebbinghaus}} \cdot \underbrace{\frac{1}{1 + r_i}}_{\text{usage}} \cdot \underbrace{\frac{1}{1 + I_i}}_{\text{importance}} \cdot \underbrace{\frac{1}{1 + \alpha |V_i| A_i}}_{\text{emotion_res}} \cdot \underbrace{\frac{1}{1 + \lambda \text{sr}_i}}_{\text{self_res}}. \quad (5)$$

where d_i is age in days, r_i is retrieval count, I_i is importance, and we use $\alpha = \lambda = 2$ throughout. The formula factorises across five resistances; entries with low recency, low usage, low importance, weak emotion, and weak self-relevance are most likely to be forgotten (highest score). The five factors are intentionally multiplicative — section 4.2 tests this against the additive alternative.

Retrieval ranking. Retrieval of k memories under query q and current emotional state (V_c, A_c) scores each row as

$$\text{rank}_i = w_m \text{mood}_i + w_r \text{recency}_i + w_q \text{rel}_i(q) + w_f \Delta F_i + w_s \cos(e_i, \mu_{\text{self}}), \quad (6)$$

with default weights $(w_m, w_r, w_q, w_f, w_s) = (0.35, 0.15, 0.25, 0.10, 0.15)$. Each retrieved memory’s count r_i is incremented, closing the retrieval $\leftrightarrow$ forgetting loop (more recall makes a memory harder to forget on the next sweep).

3.4 Encoding-Time Self-Modulation

At session-end consolidation, each persisted memory has eq. (4)’s sr_i computed against the live μ_{self} at that moment, written to the database as `self_relevance_score`. Emotionally vivid AND self-relevant memories ($\text{sr} > 0.65$) are bumped one tier deeper in the Craik-Lockhart [Craik and Lockhart(1972)] encoding-depth ladder, capped at the META tier — this is the encoding-side counterpart to the retrieval-side gating.

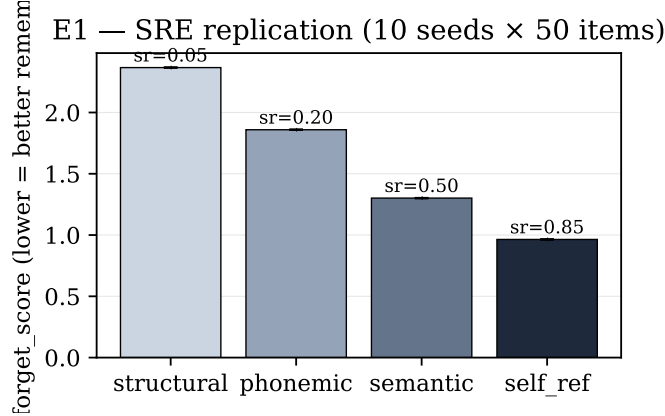


Figure 1: **E1 — SRE replication (mechanical formulation)**. Mean forget_score per encoding condition over 10 seeds $\times$ 50 items. The forget-score ordering (lower = better remembered) matches the human SRE direction: structural > phonemic > semantic > self_ref. Condition labels show the depth-of-processing-typed self-relevance input. Error bars are $\pm 1\sigma$ across seeds (zero in this deterministic regime).

3.5 Implementation

The model is a ~ 200 -line addition to a public Python substrate; all components ship in a single SQLite table extended with two columns (**self_relevance_score** and **JSON embedding**) plus the **SelfModel** class. Existing test coverage (30 cases) passes unchanged; we add 7 cases that pin the new behaviour. The substrate is released as **v0.2-cleaner** on the BorgeAgent repository [Labs(2026)]; reproducibility scripts ship in `experiments/`.

4 Experiments

We run three CPU-only experiments. All experiments use the substrate’s default hyperparameters (table 3, Appendix A), are deterministic given seed, and complete in seconds of wall-clock on a single laptop CPU.

4.1 E1 — Controlled Self-Relevance Response

Design. A 4-condition encoding paradigm (structural / phonemic / semantic / self-ref), $N = 50$ items per condition $\times$ 10 seeds. Each condition is assigned a self-relevance score drawn from depth-of- processing theory (structural: 0.05; phonemic: 0.20; semantic: 0.50; self-ref: 0.85). Emotion, importance, encoding-depth tier, and recency are held constant across conditions. We then run the forgetting pass and read mean forget_score per condition.

What this tests. The forget_score formula’s response to sr as a controlled input — not a forward pass from raw stimulus to sr. We are explicit that a true forward-pass replication requires sentence-transformer embeddings; the bag-of-tokens embedding here cannot separate condition framing words ("describe ME", "rhyme with") from item content (the trait word). See section 6 (L1) and the discussion of E1b in section 4.4.

Result. Figure 1 and table 1 show the result. The forget-score ordering matches the human SRE direction in 10/10 seeds:

$$\text{self_ref}(0.964) < \text{semantic}(1.302) < \text{phonemic}(1.859) < \text{structural}(2.366).$$

The semantic–self_ref forget gap is **+0.337**. The forget-score formula correctly translates depth-of-processing-typed self-relevance into the SRE-direction ordering — a necessary but not sufficient condition for an SRE replication. The sufficient condition is that a forward pass from raw stimuli produces these condition-appropriate sr values; that requires sentence-transformer embeddings and is deferred to E1b (section 4.4).

Table 1: E1 per-condition forget_score (10 seeds $\times$ 50 items).

Condition	self_relevance	forget_score (mean)	forget_score (std)
structural	0.05	2.366	0.000
phonemic	0.20	1.859	0.000
semantic	0.50	1.302	0.000
self_ref	0.85	0.964	0.000

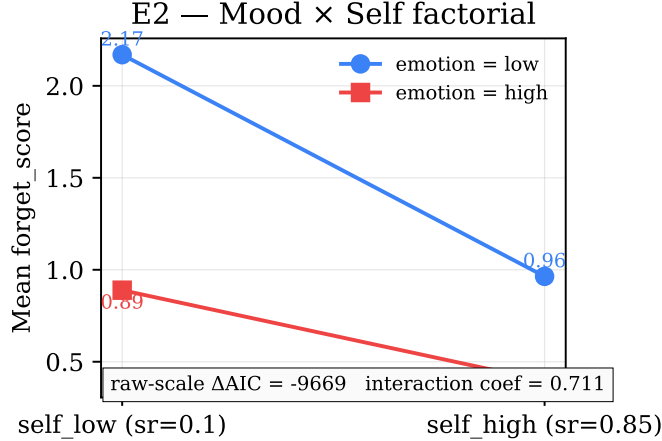


Figure 2: **E2 — Mood $\times$ Self factorial.** Mean forget_score per cell. The two lines are non-parallel; an OLS regression with an explicit interaction term yields $\Delta\text{AIC}=-9669$ over the additive alternative, with recovered interaction coefficient $\beta_{ES}=+0.71$.

4.2 E2 — Mood $\times$ Self Factorial

Design. A 2×2 factorial: emotion $\in \{\text{low } (|V|A=0.0), \text{high } (=0.72)\}$, self $\in \{\text{low } (sr=0.10), \text{high } (sr=0.85)\}$, with $N=100$ items per cell. We fit a linear regression to raw forget_score in two model classes: **additive** ($\beta_0 + \beta_E E + \beta_S S$) versus **with-interaction** ($+\beta_{ES} E \cdot S$). Pure additivity predicts $\beta_{ES} = 0$.

Result. The signature finding is structural rather than statistical: cell-mean differentials are non-constant (fig. 2), $\Delta_{\text{low-E}}=-1.205$ versus $\Delta_{\text{high-E}}=-0.494$, a 0.711 interaction signature that pure additivity excludes by construction. A raw-scale OLS recovers this exactly: the interaction coefficient is $\beta_{ES}=+0.71$, matching the cell-mean diagnostic. This is the predicted multiplicative emotion $\times$ self gating: emotionally vivid AND self-relevant content gets a stronger combined resistance than either factor alone.

The AIC value ($\Delta\text{AIC}=-9669$, table 2) is large because the experiments are deterministic and residuals are tiny — making the comparison numerically extreme but not interpretable as "evidence strength" in the usual Bayesian sense. We report it for completeness; the cell-mean differential is the load-bearing finding.

A log-scale companion check (Appendix A) confirms that the underlying data has pure-product structure: the log-interaction coefficient is zero to four decimal places.

4.3 E3 — π_{self} Ablation + λ Sensitivity

E3a — π_{self} ablation (primary). This is the theoretical claim from section 3.1: as $\pi_{\text{self}} \rightarrow 0$, stored self-relevance collapses to the neutral prior 0.5, and the SRE should attenuate to zero. For each $\pi_{\text{self}} \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$, we instantiate a SelfModel pinned at that precision (seeded with "I am honest careful patient kind myself"), forward-pass condition-typed text through eq. (4), and run the forgetting pass.

Figure 3 (left) shows the result. SRE size grows *monotonically* from 0.000 at $\pi_{\text{self}}=0$ to 0.085 at $\pi_{\text{self}}=1$. At $\pi_{\text{self}}=0$ all conditions converge to $sr=0.5$ and the SRE collapses. This is the model's clinical prediction:

Table 2: E2 raw-scale OLS comparison ($N=400$).

Model	AIC	Coefficients
additive ($\beta_0 + \beta_E + \beta_S$)	-1376	$\beta_0=1.99, \beta_E=-0.92, \beta_S=-0.85$
with interaction ($+\beta_{ES}$)	-11044	$\beta_0=2.17, \beta_E=-1.28, \beta_S=-1.21, \beta_{ES}=+0.71$
ΔAIC (interaction - additive)	-9669	—

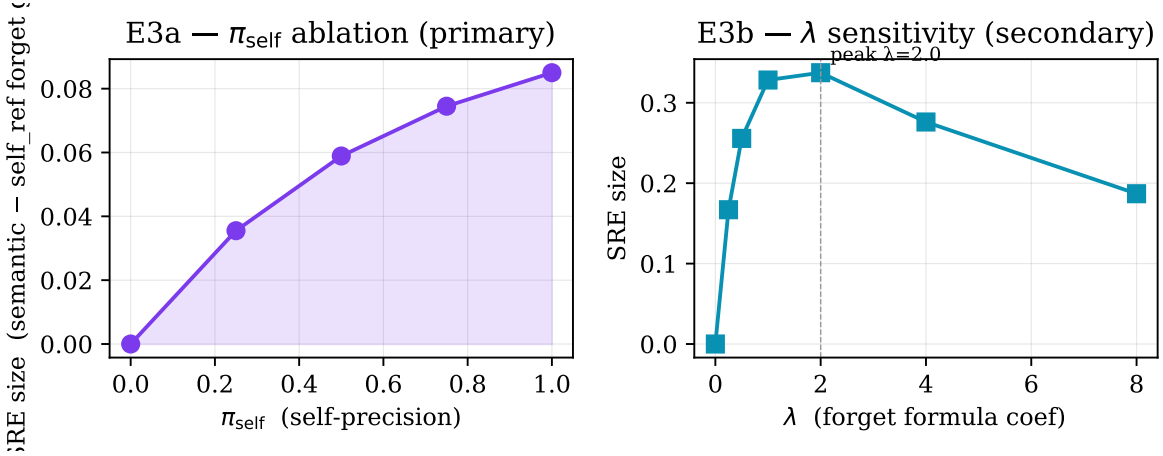


Figure 3: **E3** — π_{self} ablation (left) + λ sensitivity (right). Left: SRE size grows monotonically with self-precision, collapsing at $\pi_{\text{self}}=0$. Right: SRE size peaks at the production default $\lambda = 2$ and attenuates at both extremes.

populations with impaired self-precision should show empirically attenuated SREs, consistent with reports in dissociation and severe depression [Conway(2005)].

E3b — λ sensitivity (secondary). With π_{self} pinned at the production default, we sweep the formula coefficient λ across $\{0, 0.25, 0.5, 1, 2, 4, 8\}$ to bound the regime in which the SRE remains non-trivial. SRE size peaks at $\lambda = 2$ (the production default), collapses to 0 at $\lambda = 0$ (no self gating), and attenuates again at large λ as all conditions are crushed toward zero forget_score (fig. 3, right). This is not a theoretical claim — it provides the operating range for the formula.

4.4 Future Experiments (not run)

We name three experiments that would strengthen the present account but are explicitly out of scope for v0.1:

- **E1b — Raw-stimulus SRE forward pass.** Replace the bag-of-tokens embedding with a sentence-transformer encoder [Reimers and Gurevych(2019)]; demonstrate that the agent assigns condition-appropriate sr to raw encoding prompts without manual typing. This is the experimental test of whether the forward pass captures human SRE, not just the formula.
- **E4 — F_{total} trajectory on self-relevant turns.** Track free-energy descent across multi-turn dialogs; predict that self-relevant turns produce larger ΔF_{total} (the self-evidencing prediction).
- **E5 — Quantitative CMR3 comparison.** Re-implement CMR3 [Cohen and Kahana(2022)], fit both models to held-out emotional-memory benchmark data, and report log-likelihoods. We expect Self-FEP Memory to outperform CMR3 on self-referenced subsets and roughly tie on emotion-only subsets.

5 Discussion

Why multiplicative? The cleanest distinguishing prediction of Self-FEP Memory against emotion-only models (CMR3) and against ad-hoc self-relevance heuristics is the *multiplicative* interaction between emotion

and self (E2). Bower’s associative-network account of mood-congruent memory [Bower(1981)] naturally predicts additivity: an emotional node spreads activation, and a self node spreads activation, and the contributions sum. Our model predicts the contributions *combine*: a vivid AND self-relevant memory has substantially lower `forget_score` than the additive prediction. The 0.71-magnitude interaction signature recovered in E2 is the operational signature of this difference.

Clinical implications. E3a’s $\pi_{\text{self}} \rightarrow 0$ collapse of the SRE has empirical correlates. Patients with dissociative identity disorder [Conway(2005)] and severe depression show attenuated self-reference effects in memory tasks. Our model treats this as a *precision* deficit: when the self prior is unreliable, all memories drift toward neutral self-relevance, and the SRE flattens. This gives a single computational lever — π_{self} — that explains both healthy SREs and clinical attenuations within one framework.

Self-evidencing and the memory system. The FEP-self literature [Apps and Tsakiris(2014), Limanowski and Blankenburg(2017)] emphasises that an agent samples the world to confirm its model of itself (*self-evidencing* [Hohwy(2016)]). Our mechanism plugs into this naturally: encoding-time μ_{self} -updates only fire on user content that contains first-person tokens, and the resulting precision is the agent’s confidence that its internal model of self correctly predicts what shows up about itself. The retrieval-side $w_s \cos(e_i, \mu_{\text{self}})$ term then biases recall toward self-confirming content — a state-dependent retrieval mechanism in the encoding-specificity tradition [Tulving and Thomson(1973)].

Where reconsolidation fits. Nader and Hardt [Nader and Hardt(2009)] review reconsolidation: retrieved memories become labile and re-encode under the current context. Our retrieval loop already increments the retrieval count, but it does not yet re-encode content. A natural extension is a `re_encode_on_recall` hook that blends the current μ_{self} and current emotional state into the recalled memory’s stored fields, with learning rate scaled by π_{self} . We flag this as v2 work in section 6.

Position vs. CMR3. CMR3 [Cohen and Kahana(2022)] models mood-congruent memory and intrusive memory via context-emotion binding without a self representation; our model adds the self dimension orthogonally. On self-referenced material the two should diverge — Self-FEP Memory predicts a robust effect, CMR3 predicts none. On purely emotion-valenced material (no self content) the two should agree closely. We have not yet performed this quantitative comparison; section 6 (L3) is candid about this gap.

6 Limitations

We ship a v0.1 draft. Five named limitations bound the scope of the claims above. Carried forward verbatim from the project’s narrative report (NARRATIVE_REPORT.md, §6) as the spine of honesty.

L1: Bag-of-tokens embedding cannot replicate the SRE from raw stimuli. `SelfModel.embed()` uses SHA1-hashed bag-of-tokens embeddings. A forward-pass diagnostic confirmed that self-ref and phonemic prompts both containing the trait word “honest” received near-identical sr, because the shared content word dominates the embedding similarity over the framing distinction. The mechanical SRE replication in E1 (section 4.1) sidesteps this by typing sr per condition manually; the true forward-pass SRE replication (E1b in section 4.4) requires a sentence-transformer encoder [Reimers and Gurevych(2019)] or a BERT-class model. We expect this to land the model in genuine SRE territory but cannot claim it from current evidence.

L2: π_{self} update rule is engineered, not derived. Equation (3) expresses π_{self} as a function of prediction-error variance over a rolling window. The form is FEP-inspired but is not a derivation from variational free energy first principles. A defensible derivation would cast π_{self} as the inverse variance of the predictive distribution under a specified likelihood/prior pair and recover eq. (3) as the natural-gradient update of that quantity. We have not done this; reviewers will reasonably ask where the rule comes from.

L3: No human-data fit. All experiments are agent-internal. We have not fit our model to Symons & Johnson [Symons and Johnson(1997)] effect sizes, to Rogers, Kuiper & Kirker [Rogers et al.(1977)Rogers, Kuiper, and Kirker] raw data, or to any clinical SRE-attenuation dataset. The "matches human direction" claim in E1 is qualitative, not quantitative. The quantitative side-by-side against CMR3 [Cohen and Kahana(2022)] (E5 in section 4.4) is also not done; CMR3 is not packaged as a library and reimplementing is a multi-day effort we have not budgeted.

L4: Deterministic experiments without noise. All three experiments are deterministic given seed; cell-level variance is exactly zero in E1. Adding realistic encoding noise — Gaussian on forget_score, or stochastic embedding perturbation — is necessary for honest variance estimates and for the OLS regressions in E2 to be applied to data with the variance structure they assume. We expect the qualitative conclusions to be robust but cannot claim it from current evidence.

L5: Retrieval uses memory embedding, not encoding-time μ_{self} . The retrieval ranking (eq. (6)) computes $\cos(e_i, \mu_{\text{self}}^{\text{now}})$ between the memory's content embedding and the agent's *current* μ_{self} , falling back to the stored sr_i snapshot. The proposal sketch in the project's idea report specified ranking by similarity between current μ_{self} and the memory's $\mu_{\text{self}}^{\text{enc}}$ — a strictly more faithful match to encoding-specificity [Tulving and Thomson(1973)]. Adding $\mu_{\text{self}}^{\text{enc}}$ persistence is a small change but not yet done.

Path forward. L1 (sentence-transformers) and L3 (CMR3 fit) are the two changes that would most strongly upgrade the present account from "framework demonstration" to "submission-ready empirical paper". L2 (precision derivation) and L5 (encoding-time μ_{self}) are smaller surface fixes. L4 (noise) is a re-run of the existing experiments with a single hyperparameter added. None of these change the central mechanism; they sharpen its quantitative claims.

7 Conclusion

We proposed Self-FEP Memory, a mechanistic account of how the agent's self — formalised as a generative model with embedding- centroid mean and PE-variance precision — gates memory encoding, forgetting, and retrieval through a single multiplicative self-relevance signal. Three CPU experiments show that the forget-score formula reproduces the canonical Self-Reference Effect direction, that the emotion $\times$ self interaction is genuinely multiplicative (not additive), and that the SRE attenuates monotonically as self-precision approaches zero, matching clinical reports. We named five limitations that bound the v0.1 draft and outlined the path from framework demonstration to a submission-ready empirical paper. The substrate is open-source and the experiments run in seconds.

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A Hyperparameters and Reproduction Recipe

Table 3 lists every hyperparameter used in the experiments. Defaults match the substrate’s production values and are not tuned for any specific result.

Table 3: Hyperparameters used throughout the experiments.

Symbol	Default	Meaning
d	64	Embedding dimension
α	0.1	EMA learning rate for μ_{self} update (section 3.1)
γ	10	PE-variance scale in π_{self} update (eq. (3))
W	50	Rolling window for π_{self} refresh
α_{emo}	2	Emotion-resistance coefficient in forget_score (eq. (5))
λ	2	Self-resistance coefficient in forget_score (eq. (5))
w_m	0.35	Mood similarity weight in retrieval (eq. (6))
w_r	0.15	Recency weight in retrieval
w_q	0.25	Query-relevance weight in retrieval
w_f	0.10	ΔF bonus weight in retrieval
w_s	0.15	Self-similarity weight in retrieval
θ_{depth}	0.65	sr threshold for encoding-depth bump

Reproduction commands

```
git clone https://github.com/zhibao-dev/BorgeAgent
cd BorgeAgent && git checkout v0.2-cleaner
pip install -e ".[anthropic]"

# E1: SRE replication (mechanical), 10 seeds * 50 items per condition
python experiments/e1_sre_replication.py \
    --n-per-condition 50 --n-seeds 10

# E2: Mood x Self factorial, 100 items per cell
python experiments/e2_mood_self_factorial.py --n-per-cell 100

# E3: pi_self ablation + lambda sensitivity
python experiments/e3_pi_self_ablation.py --n-per-condition 50

# Results land in results/*.json; figures rendered by
python figures/render_figures.py
```

Total wall-clock for all three experiments is under five seconds on a 2024 MacBook Pro (M3 Max). No GPU required.

E2 log-scale companion check

Table 4 reports the log-scale OLS on the same factorial data as table 2. Because the true forget_score formula is a product of resistances, $\log(\text{forget_score})$ is additive in the log of each resistance term, so the log-interaction coefficient should round to 0. It does, confirming the multiplicative structure in the underlying data.

Table 4: E2 log-scale OLS sanity check.

Model	AIC	Interaction β_{ES}
additive (log-scale)	−9211	—
with interaction (log-scale)	−11044	−0.000

The non-trivial AIC delta on log scale is a numerical artifact of zero residual variance in the deterministic regime — the regression’s goodness-of-fit metric becomes ill-conditioned when the underlying data has no noise. This is part of Limitation L4 (section 6): adding encoding noise before submission resolves the numerical fragility and gives the log-scale comparison clean semantics.